

**Language Doesn't Matter: Evaluating the Efficacy of Large Language Models in Analyzing
Sentiment in French Textual Data**

Laurence-Olivier M. Foisy, Camille Pelletier, Étienne Proulx, and Yannick Dufresne

Département de science politique, Université Laval

Author Note

Correspondence concerning this article should be addressed to Laurence-Olivier M. Foisy,
Département de science politique, Université Laval, 2325, rue de l'Université, Québec, Québec
G1V 0A6, Canada, Email: mail@mfoisy.com

Abstract

Researchers who analyze non-English text with large language models (LLMs) inherit two rules of thumb. The first is to prompt in English, because these models learned mostly from English. The second is to translate the text into English, as dictionary methods long required. The evidence for both rules is contested. This article tests them on French. Twelve open- and closed-weight LLMs scored 200 manually annotated French news sentences under three conditions: French prompts on French text, English prompts on French text, and English prompts on machine-translated text. The English and French Lexicoder dictionaries served as benchmarks. Prompt language made no meaningful difference, and equivalence tests rule out any substantial advantage for English prompts. Translation cut both ways. It improved the dictionary, but it lowered the LLMs' polarity classification and, relative to the fully French workflow, their correlation with human judgments. The best models reached $r = 0.81$. Every model from about 8B parameters upward correlated with human judgments more strongly than either dictionary, at a cost of a few cents per thousand sentences. A 1B model could not follow the task. For sentiment analysis of French text, a validated general-purpose LLM now measures better than the Lexicoder dictionary, which remains useful as a transparent baseline. Prompt language is a matter of convenience, and the text is best analyzed in the original.

Keywords: Sentiment Analysis, Large Language Models, Machine Translation, Prompt Language, French Language, Open-Weight Models, Computational Social Science

Language Doesn't Matter: Evaluating the Efficacy of Large Language Models in Analyzing Sentiment in French Textual Data

Introduction

Computational text analysis in the social sciences grew up around English. Its tools, from sentiment dictionaries to topic models, were built for English text. This dominance has been criticized for depressing comparative research and for excluding scholarly communities that work in other languages ([Baden et al., 2022](#)). Comparativists coped by translating. Machine translation made it possible to convert multilingual corpora into English and to apply English-language tools ([De Vries et al., 2018](#); [Lucas et al., 2015](#)). Large language models (LLMs) promise to loosen this constraint. Used without task-specific training, they match or exceed crowd workers and expert coders on common annotation tasks ([Gilardi et al., 2023](#); [Törnberg, 2025](#)). They also do so, with some loss, in languages other than English ([Heseltine & Clemm Von Hohenberg, 2024](#); [Rathje et al., 2024](#)).

Yet researchers who use LLMs on non-English text often keep two reflexes from the English-centered era. The first is to prompt in English. The justification is that these models are trained mostly on English ([Touvron et al., 2023](#)) and that English instructions have been found to work better even for non-English inputs ([Lai et al., 2023](#); [X. V. Lin et al., 2022](#)). The second is to translate the text into English before analysis. This practice comes from dictionary and bag-of-words methods, and it reappears as “translate-test” with LLMs ([Etxaniz et al., 2024](#)). Whether either reflex helps, harms, or makes no difference is unsettled. The evidence comes mostly from reasoning and classification benchmarks in natural language processing, often with older models and lower-resourced languages. The social science evidence is thin.

This article tests both reflexes on sentiment analysis, a measurement task central to political communication research ([van Atteveldt et al., 2021](#); [Young & Soroka, 2012](#)). Sentiment is a demanding test for translation, because affective meaning is precisely what translation can alter ([Mohammad et al., 2016](#)). The study uses 200 manually annotated sentences from French news coverage. It compares twelve open- and closed-weight LLMs under three conditions: French

prompts on French text, English prompts on French text, and English prompts on machine-translated text. The English and French versions of the Lexicoder Sentiment Dictionary serve as benchmarks. Equivalence tests assess the absence of a prompt-language effect directly, rather than inferring it from non-significance.

The results speak to both reflexes. Prompt language made no meaningful difference. Translation cut both ways: it improved the dictionary but hurt the LLMs. The translation strategy that served English-bound tools does not carry over to models that can read the source language. The article also answers a practical question: is a researcher working on French text still better served by a dictionary? On this evidence, no. Every model from about 8B parameters upward outperformed both the native French dictionary and the translated English one on correlation with human judgments, open-weight models included, for a few cents per thousand sentences. The dictionary retains transparency and stability. These make it a baseline rather than a primary measure.

Literature Review

LLMs as Measurement Instruments in the Social Sciences

Zero-shot LLM annotation has moved quickly from curiosity to candidate standard. ChatGPT outperformed crowd workers on relevance, stance, topic, and frame detection ([Gilardi et al., 2023](#)). GPT-4 outperformed expert coders and supervised classifiers at identifying the party of politicians from their social media posts in eleven countries ([Törnberg, 2025](#)). GPT-4 also closely approximated expert coding of negativity and sentiment in tweets from the United States, Chile, Germany, and Italy ([Heseltine & Clemm Von Hohenberg, 2024](#)). These results have fed broad claims that generative AI could transform computational social science ([Bail, 2024](#); [Ziems et al., 2024](#)), chiefly by removing the need for the thousands of annotated examples that supervised learning requires ([Chae & Davidson, 2026](#); [Laurer et al., 2024](#)).

A more cautious literature qualifies these claims on three fronts. First, performance is uneven. Across 25 English benchmarks, zero-shot LLMs did not outperform the best fine-tuned models on classification tasks ([Ziems et al., 2024](#)). Across 27 annotation tasks, their performance

was “highly contingent on both the dataset and the type of annotation task” (Pangakis et al., 2023). Validation against human coding therefore remains indispensable (Grimmer & Stewart, 2013; H. Lin & Zhang, 2026), and it is only as informative as the human gold standard is reliable (Song et al., 2020). Second, established tools retain advantages. Dictionaries generalize across domains and preserve gradations of sentiment (Van Der Veen & Bleich, 2025). However, off-the-shelf dictionaries validate poorly against human judgment (van Atteveldt et al., 2021), and transformer-based models outperform them in German political and cultural texts (Klähn et al., 2025; Widmann & Wich, 2023). A domain-specific multilingual classifier can also match GPT-4o at a fraction of the cost (Mochtak et al., 2025). Third, reliance on proprietary models threatens reproducibility (Ollion et al., 2024; Spirling, 2023). This has prompted calls to justify their use (Palmer et al., 2023) and evidence that open-weight models can be competitive alternatives (Alizadeh et al., 2025; Klähn et al., 2025).

These debates leave the question of language largely implicit. The broadest evaluations rely on English benchmarks (Ziems et al., 2024). Studies that extend to other languages report somewhat lower performance outside English (Heseltine & Clemm Von Hohenberg, 2024; Törnberg, 2025), but they do not examine how researchers should interact with models in those languages. The most extensive multilingual evaluation in the social sciences covers 12 languages, yet it compared GPT with dictionary methods on English headlines only (Rathje et al., 2024). Researchers working in other languages must make two choices: the language of the prompt and the language of the text. Both have been debated elsewhere.

Prompting in English: Pivot Language or Multilingual Competence?

The case for prompting in English starts from two observations. LLMs are trained predominantly on English (Touvron et al., 2023), and they perform less well in other languages, especially lower-resourced ones (Ahuja et al., 2023; Lai et al., 2023). A mechanistic account suggests why. In Llama-2 models, intermediate representations pass through an abstract “concept space” that lies closer to English than to other languages (Wendler et al., 2024). If models effectively reason through English, instructions written in English should be understood best.

Several findings support this pivot-language view. ChatGPT performed better with English prompts even when the task and input were in other languages (Lai et al., 2023). English templates remedied the unexpectedly low accuracy of prompts handcrafted by native speakers (X. V. Lin et al., 2022). Models fine-tuned on English prompts performed best with English prompts even on non-English samples (Muennighoff et al., 2023). Asking models to restate a non-English request in English improved their multilingual performance (Huang et al., 2023). English instructions outperformed Arabic ones on average across eleven Arabic classification tasks (Kmainasi et al., 2025).

A competing view emphasizes multilingual competence. Large models show strong multilingual reasoning abilities even in underrepresented languages (Shi et al., 2022). For capable models, the language of the prompt appears to matter little. In the Arabic study just cited, differences across prompt languages were minimal for the strongest model, GPT-4o (Kmainasi et al., 2025). GPT-4 Turbo's labels for Indonesian tweets under English and Indonesian prompts agreed almost perfectly (Rathje et al., 2024). Changing the prompt language alters the length and wording of generated responses, while their meaning remains largely stable (Nguyen et al., 2026). The evidence on both sides comes mostly from reasoning and classification benchmarks, older models, and lower-resourced languages. None of it tests whether prompt language affects the accuracy of a graded sentiment judgment in a well-resourced language. The two views imply competing expectations:

H1a (pivot language). English prompts yield more accurate sentiment measurement of French text than French prompts.

H1b (multilingual competence). English and French prompts yield equivalent performance on French text.

Translating the Text: Bridge or Loss?

The second reflex predates LLMs. Faced with multilingual corpora, comparative researchers translated texts into English so that English-language tools could be applied (Lucas et

al., 2015). This bridge strategy has been validated repeatedly. Machine-translated and human-translated texts yield highly similar term-document matrices and topic models (De Vries et al., 2018). Results are robust across translation services (Reber, 2019). A machine translation of the Lexicoder dictionary captured legislative conflict across European parliaments (Proksch et al., 2019). For sentiment specifically, translating non-English sentences into English and applying English methods performed as well as language-specific methods (Araújo et al., 2020), and sentiment analysis of English translations of Arabic posts was competitive with analysis of the originals (Mohammad et al., 2016). The logic has carried into the LLM era. Translating inputs into English before inference improves the performance of multilingual models (Etxaniz et al., 2024). It is often hard to beat for lower-resourced languages written in non-Latin scripts (Ahuja et al., 2023), and it appears in applied research (Venerito & Iannone, 2024).

The opposing view treats translation as a source of loss. Translation alters sentiment: affective words disappear or are replaced by words of opposite valence, ambiguous words are mistranslated, and culturally specific expressions lose their meaning (Mohammad et al., 2016). Approaches that avoid translation, such as multilingual dictionaries (Maier et al., 2022) and multilingual sentence embeddings (Licht, 2023), yield comparable or more reliable results in some settings. Crucially, the case for translation was built largely on tools that cannot read the source language, such as English dictionaries and bag-of-words models. LLMs can. For them, the supporting evidence comes from benchmarks whose non-English versions were themselves created by translation, a setting that may favor translation-based methods, as the authors note (Etxaniz et al., 2024). It does not come from text written natively in the source language. Again, the two views imply competing expectations:

H2a (translation as bridge). Analyzing English machine translations yields sentiment measurement at least as accurate as analyzing the French original.

H2b (translation as loss). Analyzing English machine translations yields less accurate sentiment measurement than analyzing the French original.

Research Questions

These debates motivate two research questions. **RQ1:** Does the language used to interact with the model affect LLM sentiment measurement of French text? This covers the language of the prompt and the language of the text through machine translation. H1a–b and H2a–b offer competing answers. **RQ2:** Can general-purpose LLMs, open- and closed-weight,¹ measure sentiment in French text well enough, without fine-tuning, to be a viable alternative to dictionary-based methods? RQ2 is descriptive. The hypotheses were derived from the literature and were not preregistered.

Data & Methods

Data Collection and Preparation

The corpus consists of 2,683 French-language news articles from 13 media sources (9 from Québec, 1 from Ontario, 3 from France). It is an exhaustive collection of the articles discussing open-source software in the Eureka database’s “Major daily newspapers (FR)” category. Articles were selected with the search query below, over the period from January 1, 1991, the release year of the Linux kernel, to January 1, 2025. The start date excludes nothing in practice: the earliest matching article was published on June 23, 1995.

```
TEXT= ("logiciel libre" | "logiciels libres" | "open source" | "open-source" | "logiciel  
open source" | "logiciels open source" | "code source ouvert" | "software libre"  
"free software" | "code source libre" | "Free Software Foundation" | "Richard  
Stallman")
```

Table 1 shows the distribution of articles by source media.

Articles were downloaded in HTML format with a Python/Selenium scraper. Publication metadata and full text were extracted with custom R functions.

¹ “Open source” traditionally implies access aligned with free and open-source software principles. In the LLM context, the term usually refers to the public release of model weights. “Open-weight” describes this condition more accurately and is used throughout.

Table 1*Summary of Collected News Articles Discussing Open Source Software by Source Media*

source_media	article_count	earliest_date	latest_date
Libération	769	1995-06-23	2024-11-04
Monde, Le	666	2001-01-09	2024-12-26
Figaro, Le	402	1998-10-21	2024-12-04
Devoir, Le	318	1998-07-20	2024-10-26
Soleil, Le (Québec, QC)	179	1998-11-08	2023-11-18
Presse, La	152	1998-03-26	2017-10-14
Journal de Montréal, Le	65	2006-12-20	2023-04-24
Journal de Québec, Le	45	2007-03-24	2024-10-19
Tribune, La (Sherbrooke, QC)	43	2000-03-29	2022-12-03
Droit, Le (Ottawa, ON)	17	1998-11-10	2014-07-17
Nouvelliste, Le (Trois-Rivières, QC)	13	2000-01-13	2022-03-05
Quotidien, Le (Saguenay, QC)	10	2008-11-01	2019-04-10
Voix de l'Est, La (Granby, QC)	4	2010-09-17	2015-02-14

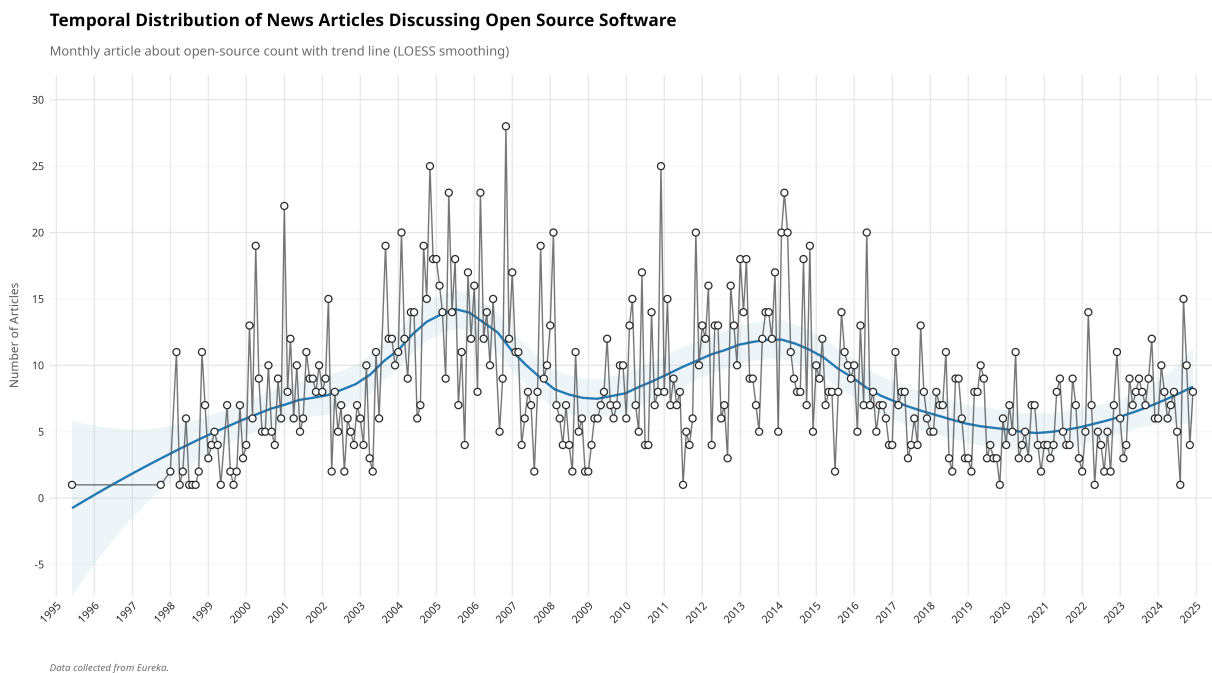
Figure 1*Temporal Distribution of News Articles Discussing Open Source Software*

Figure 1 shows the temporal distribution of articles. Coverage rose markedly in the early 2000s.

Translation

The EN→EN condition described below reproduces the translate-then-analyze strategy of comparative text analysis (De Vries et al., 2018; Lucas et al., 2015). For this condition, the French corpus was machine-translated into English with Google Translate’s document translation service. An automated R workflow processed the documents in batches and preserved document IDs, so that the French and English versions remain aligned.²

Manual Annotation

A validation benchmark of 200 sentences was randomly sampled from the corpus. Each sentence was machine-translated into English with Google Translate through the `polyglotR` package. This sample size falls within the 100 to 300 units generally recommended for validating automated sentiment measures against a manually coded gold standard (van Atteveldt et al., 2021). It yields 95% confidence intervals of roughly ± 0.05 around a correlation of 0.80. Three independent coders rated the sentiment of each sentence on a scale from -1 (very negative) to 1 (very positive), using a custom Shiny application. The three ratings were averaged. This average is the ground truth against which both dictionary and LLM methods are evaluated.

Agreement between individual coders was moderate. Krippendorff’s alpha (interval) was 0.597 (95% bootstrap CI [0.516, 0.656]). The mean pairwise correlation between coders was 0.687. All three coders assigned the same sign (negative, zero, or positive) to only 42.5% of sentences. Two features of the task account for much of this disagreement. Coders used a fine-grained scale with more than 60 distinct values, and most sentences carry mild sentiment, so small differences often cross zero. The ground truth averages three coders and is therefore more reliable than any single coder: the Spearman-Brown projection from the mean pairwise correlation puts the reliability of the three-coder mean at 0.868. These levels still call for caution. Simulations show that low intercoder reliability and small validation sets increase the risk of misjudging automated methods (Song et al., 2020). The Limitations section returns to this point.

² See `src/21_translate_to_english.R` in the project repository.

Dictionary-Based Sentiment Analysis

Lexicon-based scoring was applied to the French original and to the English translation as a classical baseline. For French text, a customized version of the French Lexicoder Sentiment Dictionary (fr1sd) (Duval & Pétry, 2016) was applied to lowercased, stopword-removed, sentence-segmented text. Each sentence was scored as (positive terms – negative terms) / total words. For the English translation, the standard Lexicoder Sentiment Dictionary (Young & Soroka, 2012) was used, with additional handling of contractions, proper nouns, and negation (for example, “not good” scored as negative). Each sentence was scored as ((positive + negated negative) – (negative + negated positive)) / total words. The two dictionaries therefore differ in construction as well as in language. This matters when they are compared.

LLM-Based Sentiment Analysis

Twelve general-purpose LLMs were evaluated. They were selected for prominence and affordability, as realistic choices for researchers who need to analyze large corpora. Proprietary models raise reproducibility concerns, so their inclusion requires justification (Palmer et al., 2023). The three closed-weight models serve as a reference point: the studies that established LLM annotation in the social sciences relied on proprietary GPT models (Gilardi et al., 2023; Heseltine & Clemm Von Hohenberg, 2024; Rathje et al., 2024; Törnberg, 2025).

The nine open-weight models are Llama 3.2 1B, Llama 3.2 3B, Llama 3.1 8B, GPT-OSS 20B, Qwen3 32B, Llama 4 Scout, GPT-OSS 120B, Qwen3 235B-A22B, and DeepSeek V3.2. They range from 1B to 671B total parameters. The three closed-weight models are Claude Haiku 4.5 (Anthropic), Gemini 3.5 Flash (Google), and GPT-5.6 Luna (OpenAI). This lineup updates an initial 2024-2025 evaluation of 11 comparable models with an identical design. Several models from that evaluation, including GPT-4o and Gemini 2.0 Flash, were deprecated by their providers within eighteen months. The smallest model, Llama 3.2 1B, rarely followed the instruction to answer with a number. About one call in 23 returned a parseable score. Even after retries, it produced a usable score for only 58 to 74 of the 200 sentences, depending on the condition. It was retained deliberately. Models of around 8B parameters can handle the task, whereas a 1B model

appears too small to follow the instructions reliably. This marks a practical lower bound on model size. Its metrics rest on a reduced and non-random subset of sentences and are not directly comparable to those of the other models.

Each model was tested under three conditions: French prompts on French text (FR→FR), English prompts on French text (EN→FR), and English prompts on the machine-translated English text (EN→EN). The first two conditions isolate the effect of prompt language. The last two isolate the effect of text language.

All prompts followed the same structure: a task definition, a five-point sentiment scale from -1.0 to 1.0 with anchor descriptions and intermediate values allowed, an instruction to consider cultural and linguistic nuance, and a constraint to respond with a numerical value only. French prompts were direct translations of the English prompts. The English-prompt/French-text version is shown below.³

```
en_prompt_fr_text <- function(text) {  
  paste0("Please analyze the sentiment of the following French text and provide a single  
    numerical rating according to this scale:  
Sentiment Scale:  
-1.0: Strong negative sentiment - highly critical, hostile, or pessimistic content  
-0.5: Moderate negative sentiment - somewhat negative, disapproving, or concerned  
    content  
0.0: Neutral sentiment - factual, balanced, or neither positive nor negative content  
0.5: Moderate positive sentiment - somewhat positive, approving, or optimistic content  
1.0: Strong positive sentiment - highly supportive, enthusiastic, or optimistic content  
You may also use values between these points for intermediate sentiment levels  
(e.g., -0.7, 0.3).  
Important instructions:  
1. First, carefully read and understand the text, considering cultural and linguistic  
    nuances in French.  
2. Analyze the emotional tone, word choice, and overall message.  
3. Respond ONLY with a single numerical value between -1.0 and 1.0 that best  
    represents the sentiment.  
4. Do not include ANY explanations, analysis, or additional text in your response.
```

³ The French-prompt/French-text version is identical except for language. See `src/93_prompts.R` in the project repository for the exact prompts, and `src/40_prompt.R` for the full evaluation pipeline, including model initialization, retry logic, rate limiting, and checkpointing.

```
Here is the text to analyze: ", text)
}
```

Each model was queried through the `ellmer` R package. The three closed-weight models were accessed through their vendors' APIs, and the nine open-weight models through OpenRouter. Responses were capped at 400 tokens. OpenRouter can serve the same model through several providers at different numerical precisions. Each open-weight model was therefore pinned to a single named provider endpoint, at the highest precision offered, with fallback to other providers disabled.⁴ Gemini 3.5 Flash was queried with its default settings. Under these settings it generates reasoning tokens before answering: 354 output tokens per call on average, against 7 for Claude Haiku 4.5. Each sentence was scored three times per model and condition (21,600 scores in total), and the mean of the three runs was taken as the final score. Despite explicit instructions to respond with a bare number, models frequently returned formatted text, such as "Sentiment: 0.5". These responses were parsed with a pattern-matching function, and only values within the -1.0 to 1.0 range were retained.

To check whether the sentence-level findings hold at the scale of a real corpus, the best-performing model on the benchmark, GPT-5.6 Luna, also scored all 2,683 full articles under the FR→FR and EN→FR conditions, with the same prompts and settings.⁵ These scores are compared with the French Lexicoder dictionary in the Appendix (Figure 5).

Validation and Performance Evaluation

Performance was assessed with four metrics: Pearson correlation between each model's continuous predictions and the ground truth; mean absolute error (MAE); weighted F1 on a 7-category classification, from "very negative" to "very positive"; and weighted F1 on a 3-category classification (negative, neutral, positive) that groups the 7 categories into polarity bands. Weighted rather than macro F1 is used because the manually coded categories are

⁴ Endpoint pins, and the reasoning settings used to suppress chain-of-thought in models that reason by default, are listed in `src/94_models_map.R`.

⁵ See `src/70_prompt_corpus.R`. The two passes required 5,366 API calls, 7.1 million input tokens and 187,000 output tokens, for USD 1.64 at list prices.

unevenly distributed. “Somewhat positive” alone accounts for 44% of annotated sentences, while “very negative” and “very positive” each account for 1% or less. Comparing the two classification schemes separates fine-grained discrimination (7 categories) from polarity classification (3 categories). All metrics were calculated on complete cases only.

Condition comparisons are paired by model and map onto the hypotheses. The prompt-language hypotheses compare $FR \rightarrow FR$ with $EN \rightarrow FR$, which hold the French text constant. The translation hypotheses compare $EN \rightarrow FR$ with $EN \rightarrow EN$, which hold the English prompt constant and isolate the effect of analyzing the translation. The comparison of $FR \rightarrow FR$ with $EN \rightarrow EN$ contrasts the two complete workflows a researcher might adopt. A non-significant t-test cannot by itself establish the absence of an effect. Alongside paired t-tests, two one-sided tests (TOST) therefore assess whether a difference is small enough to be treated as negligible, with an equivalence bound of ± 0.05 on each metric’s scale. These comparisons and the averages reported in the Results cover the eleven models that scored all 200 sentences (33 model-condition combinations). They exclude the dictionary baselines, which have no prompt. Llama 3.2 1B is reported separately: it scored a different, partial subset of sentences in each condition, so comparisons across conditions would compare unlike samples.

Results

Prompt Language (H1a vs. H1b)

Holding the text constant, prompt language made no detectable difference. French prompts on French text ($FR \rightarrow FR$) scored slightly higher than English prompts on the same text ($EN \rightarrow FR$) on every metric: mean correlation 0.700 vs. 0.673, MAE 0.209 vs. 0.217, 3-category F1 0.669 vs. 0.646, and 7-category F1 0.364 vs. 0.347. None of these paired differences was statistically significant (all $p > 0.12$). Equivalence tests go further. $FR \rightarrow FR$ and $EN \rightarrow FR$ were statistically equivalent within ± 0.05 on MAE ($p = 0.002$), 7-category F1 ($p = 0.034$), and 3-category F1 ($p = 0.034$). Correlation was the one metric on which equivalence could not be established. The 90% confidence interval of the difference, $[-0.032, 0.085]$, leaves room for a French-prompt advantage of up to 0.085, but not for an English-prompt advantage larger than

0.032. The results contradict H1a and support H1b on three of four metrics.

Machine Translation (H2a vs. H2b)

For the LLMs, translation carried a cost. Holding the English prompt constant, analyzing the translation instead of the original (EN→EN vs. EN→FR) lowered 3-category F1 by 0.04 (0.606 vs. 0.646, $p = 0.005$). The accompanying drop in correlation (0.647 vs. 0.673) was not statistically significant ($p = 0.24$). Differences in MAE and 7-category F1 remained within the ± 0.05 equivalence bound. Compared with the fully French workflow (FR→FR), the fully English workflow (EN→EN) lowered correlation by 0.053 ($p = 0.007$) and 3-category F1 by 0.063 ($p < 0.001$).

The dictionaries showed the opposite pattern. The English Lexicoder applied to the translation correlated with human judgments at $r = 0.548$, against 0.357 for the French Lexicoder applied to the original. The difference is 0.191 (95% bootstrap CI [0.059, 0.326]). The English dictionary also reached a higher 3-category F1 (0.588 vs. 0.557). The two dictionaries also differ in construction, so this contrast does not isolate language. Its direction nonetheless matches the bridge view. For the LLMs, the results contradict H2a and support H2b for polarity classification and, in the comparison of complete workflows, for correlation.

Viability Against Dictionary Methods (RQ2)

General-purpose LLMs, used without fine-tuning, correlated with human judgments well above both dictionaries. The best configurations exceeded $r = 0.80$. GPT-5.6 Luna led on correlation ($r = 0.810$ in FR→FR). Claude Haiku 4.5 led on 3-category F1 (0.778) and 7-category F1 (0.537), both in FR→FR. Gemini 3.5 Flash led on MAE (0.156 in EN→FR). Nine of the 33 model-condition combinations correlated with the ground truth more strongly than individual human coders agreed with the mean of the other two (mean leave-one-out $r = 0.748$, range 0.724–0.779).⁶

⁶ This comparison slightly favors the models, because the ground truth they are compared against averages three coders rather than two. Agreement between annotators is not a strict ceiling on machine performance (Richie et al., 2022).

Figure 2
Sentiment Analysis Models Correlation with Ground Truth

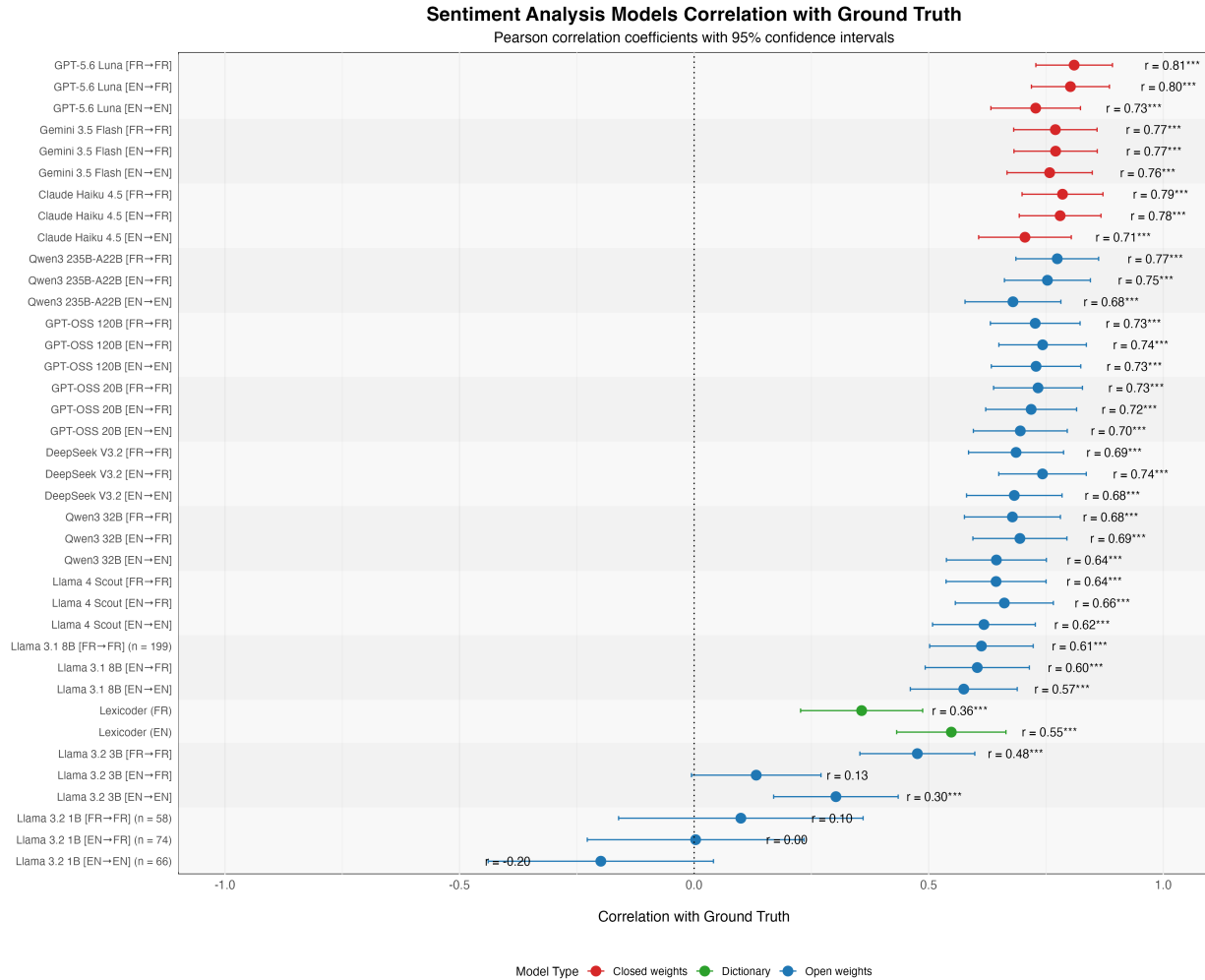


Figure 1. Cross-lingual sentiment analysis performance across model architectures. Models grouped by type (color) and sorted by correlation strength. Prompting mechanisms within clusters ordered as FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text). Error bars represent 95% confidence intervals; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Closed-weight models outperformed open-weight ones on average (mean $r = 0.768$ vs. 0.638; 3-category F1 0.689 vs. 0.622). The gap closed at the top of the open-weight range. Qwen3 235B-A22B reached $r = 0.774$ in FR→FR, above Gemini 3.5 Flash in the same condition. GPT-OSS 120B and GPT-OSS 20B averaged $r = 0.733$ and 0.716 across conditions. Performance degraded below that size. Llama 3.1 8B remained usable ($r = 0.575$ –0.612). Llama 3.2 3B was erratic ($r = 0.476$ in FR→FR but 0.132 in EN→FR). Llama 3.2 1B failed: it returned a usable score for only 58 to 74 sentences per condition, and on those its correlation with the ground truth ranged from -0.199 to 0.100. Cost was not a barrier. At list prices, scoring 1,000 sentences cost

between USD 0.02 and 0.40 for every model but one (Table B1). The exception, Gemini 3.5 Flash (USD 3.77), reflects the reasoning tokens it generates by default rather than its per-token price.⁷

Against the LLM mean of $r = 0.673$, the dictionaries trailed on correlation (0.548 and 0.357) and on 3-category F1 (0.588 and 0.557 vs. 0.640). On correlation, the comparison is one-sided: every model from 8B parameters upward exceeded both dictionaries in every condition, and only Llama 3.2 3B and 1B fell below them. On 3-category F1, the translated English dictionary outperformed Qwen3 32B in every condition, GPT-OSS 120B in EN→EN, and the two smallest Llama models. On MAE, both dictionaries (0.203 and 0.210) did slightly better than the LLM average (0.213). This reflects the dictionaries' compressed output rather than their accuracy. Their scores cluster near zero (SD = 0.065 for the English dictionary, against 0.286 for the ground truth and 0.38–0.47 for the LLMs). On a sample dominated by mildly positive sentences, this keeps absolute errors small while discriminating poorly between sentences.

Classification Granularity

All models performed substantially better on 3-category than on 7-category classification (best F1 0.778 vs. 0.537; mean 0.640 vs. 0.349; Figure 3). Models identified positive and negative sentiment most reliably (7-category F1 = 0.482 and 0.441, rising to 0.690 and 0.627 in the 3-category scheme). They struggled most with the extreme categories (very negative F1 = 0.256, very positive F1 = 0.124), which are nearly absent from the sample (two sentences and one sentence). Neutral remained the hardest 3-category label (F1 = 0.258).

⁷ No cost is reported for Llama 3.2 1B. Its endpoint returned the same input token count on every call regardless of the prompt, so its usage was not measured. It needed 22.7 API calls per usable score, against 1.0 to 1.03 for every other model, and would in any case have been the most expensive open-weight model to run.

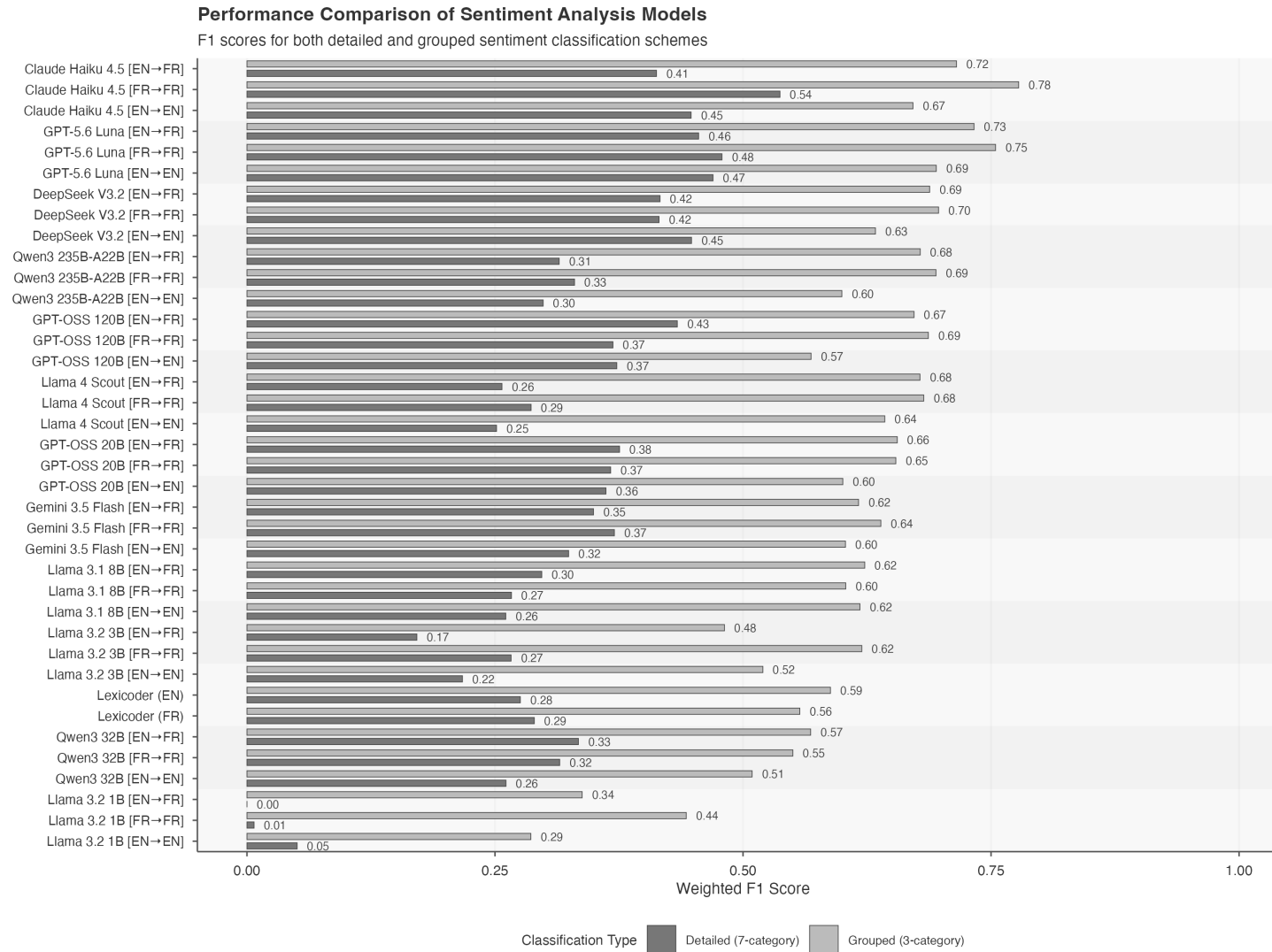


Figure 2. F1 scores for sentiment analysis across model architectures. Models grouped by F1 score performance and sorted by 3-category classification. Prompting mechanisms: EN→FR (English prompt on French text), FR→FR (French prompt on French text), and EN→EN (English prompt on translated text). Detailed classification includes 7 categories, grouped classification uses 3.

Figure 3
F1 Scores Comparison Across Models and Categories

Discussion

Prompt Language: Multilingual Competence, Within Scope Conditions

For French and current models, the results side with the multilingual-competence view. English prompts brought no advantage, and equivalence tests ruled out any English-prompt gain larger than a few hundredths on every metric. Earlier social science evidence showed that labels agree across prompt languages ([Rathje et al., 2024](#)). This study extends that finding to the accuracy of the labels against human judgment. It is also consistent with the minimal differences observed for the strongest model in Arabic ([Kmainasi et al., 2025](#)). The results do not refute the pivot-language evidence so much as delimit it. That evidence comes largely from reasoning and inference benchmarks, older or smaller models, and lower-resourced languages ([Lai et al., 2023](#); [X. V. Lin et al., 2022](#); [Muennighoff et al., 2023](#)). The representational bias toward English that it points to ([Wendler et al., 2024](#)) need not produce worse instruction-following for a simple judgment in a language as well represented as French. The smaller open-weight models behaved differently. Llama 3.2 3B correlated with human judgments at $r = 0.48$ with French prompts but 0.13 with English ones. This suggests that sensitivity to prompt language may reappear below a certain model size, although a single model cannot establish this. At the scale of the full corpus, the equivalence of prompt languages is even clearer. GPT-5.6 Luna's scores for the 2,683 articles under French and English prompts correlated at $r = 0.97$, and 94% of articles received the same sign (Figure 5).

Prompt language does carry a cost that the accuracy results do not show. The French prompt is 6% longer than the English one in characters, yet it cost 13% to 34% more input tokens, depending on the model (Table B2). The difference is a property of each model's tokenizer, not of its size or licence. Vocabularies that favor English encode English text with similar efficiency across models (1.17 to 1.25 tokens per word). They split French words into more pieces, at rates from 1.41 tokens per word (Llama 4 Scout, GPT-5.6 Luna, GPT-OSS) to 1.87 (Claude Haiku 4.5). Ahia et al. ([2023](#)) document the same disparity across many languages. The French sentences themselves cost 1.6% to 4.6% more tokens than their English translations. Output length did not

differ by prompt language. Since accuracy was equivalent, the English prompt is the cheaper of two equally good options. The saving depends on where a model's cost lies. It is 14% to 31% per 1,000 sentences for models billed mostly on input (Claude Haiku 4.5, Llama 3.x, Qwen3, DeepSeek V3.2, Llama 4 Scout). It is under 6% for GPT-5.6 Luna, the GPT-OSS models, and Gemini 3.5 Flash, whose output tokens weigh more. Translating the text first would save a few percent more, at the accuracy cost reported above.

Translation: A Bridge for Dictionaries, a Loss for LLMs

The translation results are the article's main contribution to the debate. The same machine translation that improved the dictionary lowered the LLMs' performance. This divergence reconciles the bridge and loss views rather than choosing between them. Translation helps tools that cannot read the source language, and it costs information for tools that can. The dictionary result is consistent with a long line of evidence that translating into English lets researchers use better-validated English resources ([Araújo et al., 2020](#); [De Vries et al., 2018](#); [Proksch et al., 2019](#)). LLMs already read French. For them, translation offers no such gain, and it exposes the analysis to the alterations of affective meaning that translation introduces ([Mohammad et al., 2016](#)). The finding runs against evidence that translating inputs into English improves multilingual models ([Etxaniz et al., 2024](#)). That evidence rests on reasoning and inference benchmarks whose non-English versions were produced by translation. The sentences analyzed here were written in French. The practical implication is direct: the translation step that dictionary-era workflows made necessary should not be carried into LLM-based sentiment analysis of French text.

Viability: Optimism, With Validation

On RQ2, the results side with the optimistic literature ([Gilardi et al., 2023](#); [Rathje et al., 2024](#); [Törnberg, 2025](#)) and extend it in two directions. First, the advantage of LLMs over dictionaries holds against a native-language dictionary as well as a translated one. The multilingual evidence had so far made this comparison only in English. Second, open-weight models were viable from about 8B parameters, and the largest approached the best closed-weight models. This is consistent with evidence from German ([Klähn et al., 2025](#)). Below that size,

performance became erratic, and a 1B-parameter model could not follow the task. For researchers weighing the reproducibility concerns raised by proprietary models (Ollion et al., 2024; Spirling, 2023), this makes a credible open-weight option available at low cost. Qwen3 235B-A22B and GPT-OSS 120B reached 90–95% of the best closed-weight correlation for USD 0.02–0.03 per thousand sentences. For French text, the answer to RQ2 is therefore yes. A general-purpose LLM measures sentiment better than the Lexicoder dictionary on every criterion that rewards discrimination, and the cost of running one is no longer a reason to prefer the dictionary.

The cautious literature is not dismissed. Performance varied widely across models, which reinforces the need to validate each model on each task (Pangakis et al., 2023). Two further results qualify the picture. First, the dictionaries' near-zero output gave them a slightly lower MAE than the LLMs. MAE alone can thus reward a method that barely discriminates. Second, all models performed markedly worse on 7-category classification than on 3-category polarity. These two results are compatible once ordering and calibration are distinguished. LLMs order sentences and reproduce the spread of human ratings. Their correlation with the ground truth (mean $r = 0.673$, up to 0.810) far exceeds that of the dictionaries (0.548 and 0.357), and their scores spread at least as widely as the human ratings (SD 0.38–0.47 vs. 0.286), whereas the dictionaries' scores cluster near zero (SD = 0.065 for the English dictionary). What LLMs do poorly is place scores on the right side of fixed category thresholds. They overstate extremes: 10.8% of LLM scores exceed ± 0.66 , against 1.5% of ground-truth values. A sentence rated 0.5 by humans and 0.8 by a model is correctly ordered but misclassified. The ground truth contributes to the mismatch. As an average of three coders, it is pulled toward the center, and only three sentences fall in the two extreme categories. Gradations in sentiment are one of the main arguments for dictionaries (Van Der Veen & Bleich, 2025). In this corpus, a single off-the-shelf dictionary captured them less well than the LLMs. The gap does not close with aggregation. On the full corpus, the French dictionary's article scores correlated with GPT-5.6 Luna's at $r = 0.36$, and their monthly averages at $r = 0.42$ (Figure 5). What the dictionary retains is transparency, since every score can be traced to the words that produced it, and stability over time, since a dictionary is not deprecated. These make it

a useful baseline and robustness check alongside an LLM. They are not a reason to prefer it as the primary measure.

Limitations

Several limitations qualify these findings. The benchmark is modest in size and reliability. Its 200 sentences fall within common recommendations for validating sentiment measures ([van Atteveldt et al., 2021](#)), but agreement between individual coders was moderate (Krippendorff's $\alpha = 0.597$). Simulations show that such conditions raise the risk of misjudging automated methods ([Song et al., 2020](#)). Averaging three coders yields a more reliable ground truth, but it also pulls ratings toward the center. Only two sentences fall in the very negative category and one in the very positive, so class-level F1 scores for these categories are unstable. The findings concern French only. The corpus is French-language news on a single domain, open-source software. Neither the absence of a prompt-language effect nor the cost of translation can be assumed to extend to other languages or genres. This applies especially to languages less represented in training data, where the pivot-language evidence is strongest.

Methodologically, the EN→EN condition relies on Google Translate, and translation quality was deliberately not validated. The condition reproduces what a researcher who does not read the source language can do: translate automatically and analyze the result. Its lower performance should be read as the cost of that workflow rather than of translation as such. The dictionary comparison confounds language with dictionary construction. A single standardized prompt was used across models for comparability. This may understate the performance achievable with model-specific prompting. The models tested do not include fine-tuned or domain-specific classifiers, which can rival general-purpose LLMs ([Mochtak et al., 2025](#)). Costs are reported at list prices and count cached prompt tokens at the full input rate, so they are upper bounds. Finally, this evaluation is a snapshot. Several models from an earlier version of this study were deprecated within eighteen months, and the specific rankings should be expected to age quickly.

Conclusion

Researchers analyzing French text with LLMs can set aside both inherited reflexes. The language of the prompt is a matter of convenience. English instructions brought no advantage, and equivalence tests rule out any substantial one. The language of the text is not a matter of convenience. Translating into English, the step that English-bound tools made necessary, improved the dictionary but lowered the LLMs' performance. The comparative strategy that served the dictionary era does not transfer to models that read the source language.

Two questions follow. The first is whether these results hold beyond French, particularly for languages less represented in training data, where the case for English prompts and translation is strongest. The design ports directly to such tests: the same three conditions, a manually annotated benchmark, and a different target language. The second is whether the size threshold observed here, below which open-weight models become erratic and prompt-sensitive, is a stable feature of current models.

On the practical question of whether a researcher working on French text is still better served by a dictionary, the answer here is no. A general-purpose LLM validated on a manually annotated sample measures sentiment better than the Lexicoder dictionary on every criterion but transparency. Open-weight models from about 8B parameters do so for a few cents per thousand sentences. The dictionary keeps two advantages, transparency and stability over time. These make it a baseline and a robustness check rather than the primary measure. Every model still needs to be validated on the task at hand. For researchers working with French text, the evidence settles three points: use a validated LLM, prompt in whichever language is convenient, and analyze the text in the language in which it was written.

References

- Ahia, O., Kumar, S., Gonen, H., Kasai, J., Mortensen, D., Smith, N., & Tsvetkov, Y. (2023). Do All Languages Cost the Same? Tokenization in the Era of Commercial Language Models. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 9904–9923. <https://doi.org/10.18653/v1/2023.emnlp-main.614>

- Ahuja, K., Diddee, H., Hada, R., Ochieng, M., Ramesh, K., Jain, P., Nambi, A., Ganu, T., Segal, S., Ahmed, M., Bali, K., & Sitaram, S. (2023). MEGA: Multilingual Evaluation of Generative AI. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 4232–4267. <https://doi.org/10.18653/v1/2023.emnlp-main.258>
- Alizadeh, M., Kubli, M., Samei, Z., Dehghani, S., Zahedivafa, M., Bermeo, J. D., Korobeynikova, M., & Gilardi, F. (2025). Open-source LLMs for text annotation: A practical guide for model setting and fine-tuning. *Journal of Computational Social Science*, 8(1), 17. <https://doi.org/10.1007/s42001-024-00345-9>
- Araújo, M., Pereira, A., & Benevenuto, F. (2020). A comparative study of machine translation for multilingual sentence-level sentiment analysis. *Information Sciences*, 512, 1078–1102. <https://doi.org/10.1016/j.ins.2019.10.031>
- Baden, C., Pipal, C., Schoonvelde, M., & Van Der Velden, M. A. C. G. (2022). Three Gaps in Computational Text Analysis Methods for Social Sciences: A Research Agenda. *Communication Methods and Measures*, 16(1), 1–18. <https://doi.org/10.1080/19312458.2021.2015574>
- Bail, C. A. (2024). Can Generative AI improve social science? *Proceedings of the National Academy of Sciences*, 121(21), e2314021121. <https://doi.org/10.1073/pnas.2314021121>
- Chae, Y., & Davidson, T. (2026). Large Language Models for Text Classification: From Zero-Shot Learning to Instruction-Tuning. *Sociological Methods & Research*, 55(2), 501–567. <https://doi.org/10.1177/00491241251325243>
- De Vries, E., Schoonvelde, M., & Schumacher, G. (2018). No Longer Lost in Translation: Evidence that Google Translate Works for Comparative Bag-of-Words Text Applications. *Political Analysis*, 26(4), 417–430. <https://doi.org/10.1017/pan.2018.26>
- Duval, D., & Pétry, F. (2016). L'analyse automatisée du ton médiatique : Construction et utilisation de la version française du Lexicoder Sentiment Dictionary. *Canadian Journal of Political Science/Revue Canadienne de Science Politique*, 49(2), 197–220. <https://doi.org/10.1017/S000842391600055X>

- Etzaniz, J., Azkune, G., Soroa, A., Lopez de Lacalle, O., & Artetxe, M. (2024). Do multilingual language models think better in English? *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*, 550–564.
<https://doi.org/10.18653/v1/2024.naacl-short.46>
- Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30), e2305016120. <https://doi.org/10.1073/pnas.2305016120>
- Grimmer, J., & Stewart, B. M. (2013). Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts. *Political Analysis*, 21(3), 267–297.
<https://doi.org/10.1093/pan/mps028>
- Heseltine, M., & Clemm Von Hohenberg, B. (2024). Large language models as a substitute for human experts in annotating political text. *Research & Politics*, 11(1), 20531680241236239. <https://doi.org/10.1177/20531680241236239>
- Huang, H., Tang, T., Zhang, D., Zhao, X., Song, T., Xia, Y., & Wei, F. (2023). Not All Languages Are Created Equal in LLMs: Improving Multilingual Capability by Cross-Lingual-Thought Prompting. *Findings of the Association for Computational Linguistics: EMNLP 2023*, 12365–12394.
<https://doi.org/10.18653/v1/2023.findings-emnlp.826>
- Klähn, J., Borst-Graetz, J., & Burghardt, M. (2025). From dictionaries to LLMs – an evaluation of sentiment analysis techniques for German language data. *Computational Humanities Research*, 1, e4. <https://doi.org/10.1017/chr.2025.10005>
- Kmainasi, M. B., Khan, R., Shahroor, A. E., Bendou, B., Hasanain, M., & Alam, F. (2025). Native vs non-native language prompting: A comparative analysis. *Web Information Systems Engineering – WISE 2024*. https://doi.org/10.1007/978-981-96-0576-7_30
- Lai, V., Ngo, N., Pourn Ben Veyseh, A., Man, H., Dernoncourt, F., Bui, T., & Nguyen, T. (2023). ChatGPT Beyond English: Towards a Comprehensive Evaluation of Large Language

- Models in Multilingual Learning. *Findings of the Association for Computational Linguistics: EMNLP 2023*, 13171–13189.
<https://doi.org/10.18653/v1/2023.findings-emnlp.878>
- Laurer, M., Van Atteveldt, W., Casas, A., & Welbers, K. (2024). Less Annotating, More Classifying: Addressing the Data Scarcity Issue of Supervised Machine Learning with Deep Transfer Learning and BERT-NLI. *Political Analysis*, 32(1), 84–100.
<https://doi.org/10.1017/pan.2023.20>
- Licht, H. (2023). Cross-Lingual Classification of Political Texts Using Multilingual Sentence Embeddings. *Political Analysis*, 31(3), 366–379. <https://doi.org/10.1017/pan.2022.29>
- Lin, H., & Zhang, Y. (2026). Navigating the Risks of Using Large Language Models for Text Annotation in Social Science Research. *Social Science Computer Review*, 44(3), 403–427.
<https://doi.org/10.1177/08944393251366243>
- Lin, X. V., Mihaylov, T., Artetxe, M., Wang, T., Chen, S., Simig, D., Ott, M., Goyal, N., Bhosale, S., Du, J., Pasunuru, R., Shleifer, S., Koura, P. S., Chaudhary, V., O'Horo, B., Wang, J., Zettlemoyer, L., Kozareva, Z., Diab, M., ... Li, X. (2022). Few-shot Learning with Multilingual Generative Language Models. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.
<https://doi.org/10.18653/v1/2022.emnlp-main.616>
- Lucas, C., Nielsen, R. A., Roberts, M. E., Stewart, B. M., Storer, A., & Tingley, D. (2015). Computer-Assisted Text Analysis for Comparative Politics. *Political Analysis*, 23(2), 254–277. <https://doi.org/10.1093/pan/mpu019>
- Maier, D., Baden, C., Stoltenberg, D., De Vries-Kedem, M., & Waldherr, A. (2022). Machine Translation Vs. Multilingual Dictionaries Assessing Two Strategies for the Topic Modeling of Multilingual Text Collections. *Communication Methods and Measures*, 16(1), 19–38.
<https://doi.org/10.1080/19312458.2021.1955845>
- Mochtak, M., Rupnik, P., Kuzman, T., & Ljubešić, N. (2025). Parlasent: Mapping sentiment in political discourse with large language models. *Political Research Exchange*, 7(1),

2508377. <https://doi.org/10.1080/2474736X.2025.2508377>
- Mohammad, S. M., Salameh, M., & Kiritchenko, S. (2016). How translation alters sentiment. *Journal of Artificial Intelligence Research*, 55, 95–130. <https://doi.org/10.1613/jair.4787>
- Muennighoff, N., Wang, T., Sutawika, L., Roberts, A., Biderman, S., Le Scao, T., Bari, M. S., Shen, S., Yong, Z. X., Schoelkopf, H., Tang, X., Radev, D., Aji, A. F., Almubarak, K., Albanie, S., Alyafeai, Z., Webson, A., Raff, E., & Raffel, C. (2023). Crosslingual Generalization through Multitask Finetuning. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. <https://doi.org/10.18653/v1/2023.acl-long.891>
- Nguyen, T. T. N., Tieu, M. K., Riegler, M. A., Halvorsen, P., & Nguyen, T. (2026). *Investigating the Influence of Prompt and Response Languages on LLM Content Generation*. arXiv. <https://doi.org/10.48550/arXiv.2608.26186>
- Ollion, É., Shen, R., Macanovic, A., & Chatelain, A. (2024). The dangers of using proprietary LLMs for research. *Nature Machine Intelligence*, 6(1), 4–5. <https://doi.org/10.1038/s42256-023-00783-6>
- Palmer, A., Smith, N. A., & Spirling, A. (2023). Using proprietary language models in academic research requires explicit justification. *Nature Computational Science*, 4(1), 2–3. <https://doi.org/10.1038/s43588-023-00585-1>
- Pangakis, N., Wolken, S., & Fasching, N. (2023). *Automated Annotation with Generative AI Requires Validation* (arXiv:2306.00176). arXiv. <https://doi.org/10.48550/arXiv.2306.00176>
- Proksch, S.-O., Lowe, W., Wäckerle, J., & Soroka, S. (2019). Multilingual Sentiment Analysis: A New Approach to Measuring Conflict in Legislative Speeches. *Legislative Studies Quarterly*, 44(1), 97–131. <https://doi.org/10.1111/lsq.12218>
- Rathje, S., Mirea, D.-M., Sucholutsky, I., Marjeh, R., Robertson, C. E., & Van Bavel, J. J. (2024). GPT is an effective tool for multilingual psychological text analysis. *Proceedings of the National Academy of Sciences*, 121(34), e2308950121.

<https://doi.org/10.1073/pnas.2308950121>

Reber, U. (2019). Overcoming Language Barriers: Assessing the Potential of Machine Translation and Topic Modeling for the Comparative Analysis of Multilingual Text Corpora. *Communication Methods and Measures*, 13(2), 102–125.

<https://doi.org/10.1080/19312458.2018.1555798>

Richie, R., Grover, S., & Tsui, F. (2022). Inter-annotator agreement is not the ceiling of machine learning performance: Evidence from a comprehensive set of simulations. *Proceedings of the 21st Workshop on Biomedical Language Processing*, 275–284.

<https://doi.org/10.18653/v1/2022.bionlp-1.26>

Shi, F., Suzgun, M., Freitag, M., Wang, X., Srivats, S., Vosoughi, S., Chung, H. W., Tay, Y., Ruder, S., Zhou, D., Das, D., & Wei, J. (2022). *Language Models are Multilingual Chain-of-Thought Reasoners* (arXiv:2210.03057). arXiv.

<https://doi.org/10.48550/arXiv.2210.03057>

Song, H., Tolochko, P., Eberl, J.-M., Eisele, O., Greussing, E., Heidenreich, T., Lind, F., Galyga, S., & Boomgaarden, H. G. (2020). In Validations We Trust? The Impact of Imperfect Human Annotations as a Gold Standard on the Quality of Validation of Automated Content Analysis. *Political Communication*, 37(4), 550–572.

<https://doi.org/10.1080/10584609.2020.1723752>

Spirling, A. (2023). Why open-source generative AI models are an ethical way forward for science. *Nature*, 616, 413. <https://doi.org/10.1038/d41586-023-01295-4>

Törnberg, P. (2025). Large language models outperform expert coders and supervised classifiers at annotating political social media messages. *Social Science Computer Review*, 43(6), 1181–1195. <https://doi.org/10.1177/08944393241286471>

Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., & Lample, G. (2023). *LLaMA: Open and Efficient Foundation Language Models* (arXiv:2302.13971). arXiv.

<https://arxiv.org/abs/2302.13971>

- van Atteveldt, W., van der Velden, M. A. C. G., & Boukes, M. (2021). The validity of sentiment analysis: Comparing manual annotation, crowd-coding, dictionary approaches, and machine learning algorithms. *Communication Methods and Measures*, 15(2), 121–140. <https://doi.org/10.1080/19312458.2020.1869198>
- Van Der Veen, A. M., & Bleich, E. (2025). The advantages of lexicon-based sentiment analysis in an age of machine learning. *PLOS ONE*, 20(1), e0313092. <https://doi.org/10.1371/journal.pone.0313092>
- Venerito, V., & Iannone, F. (2024). Large language model-driven sentiment analysis for facilitating fibromyalgia diagnosis. *RMD Open*, 10(2), e004367. <https://doi.org/10.1136/rmdopen-2024-004367>
- Wendler, C., Veselovsky, V., Monea, G., & West, R. (2024). Do Llamas Work in English? On the Latent Language of Multilingual Transformers. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 15366–15394. <https://doi.org/10.18653/v1/2024.acl-long.820>
- Widmann, T., & Wich, M. (2023). Creating and Comparing Dictionary, Word Embedding, and Transformer-Based Models to Measure Discrete Emotions in German Political Text. *Political Analysis*, 31(4), 626–641. <https://doi.org/10.1017/pan.2022.15>
- Young, L., & Soroka, S. (2012). Affective News: The Automated Coding of Sentiment in Political Texts. *Political Communication*, 29(2), 205–231. <https://doi.org/10.1080/10584609.2012.671234>
- Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., & Yang, D. (2024). Can Large Language Models Transform Computational Social Science? *Computational Linguistics*, 50(1), 237–291. https://doi.org/10.1162/coli_a_00502

Appendix A

Appendix B

Figure B1
Sentiment Analysis Models Mean Absolute Error

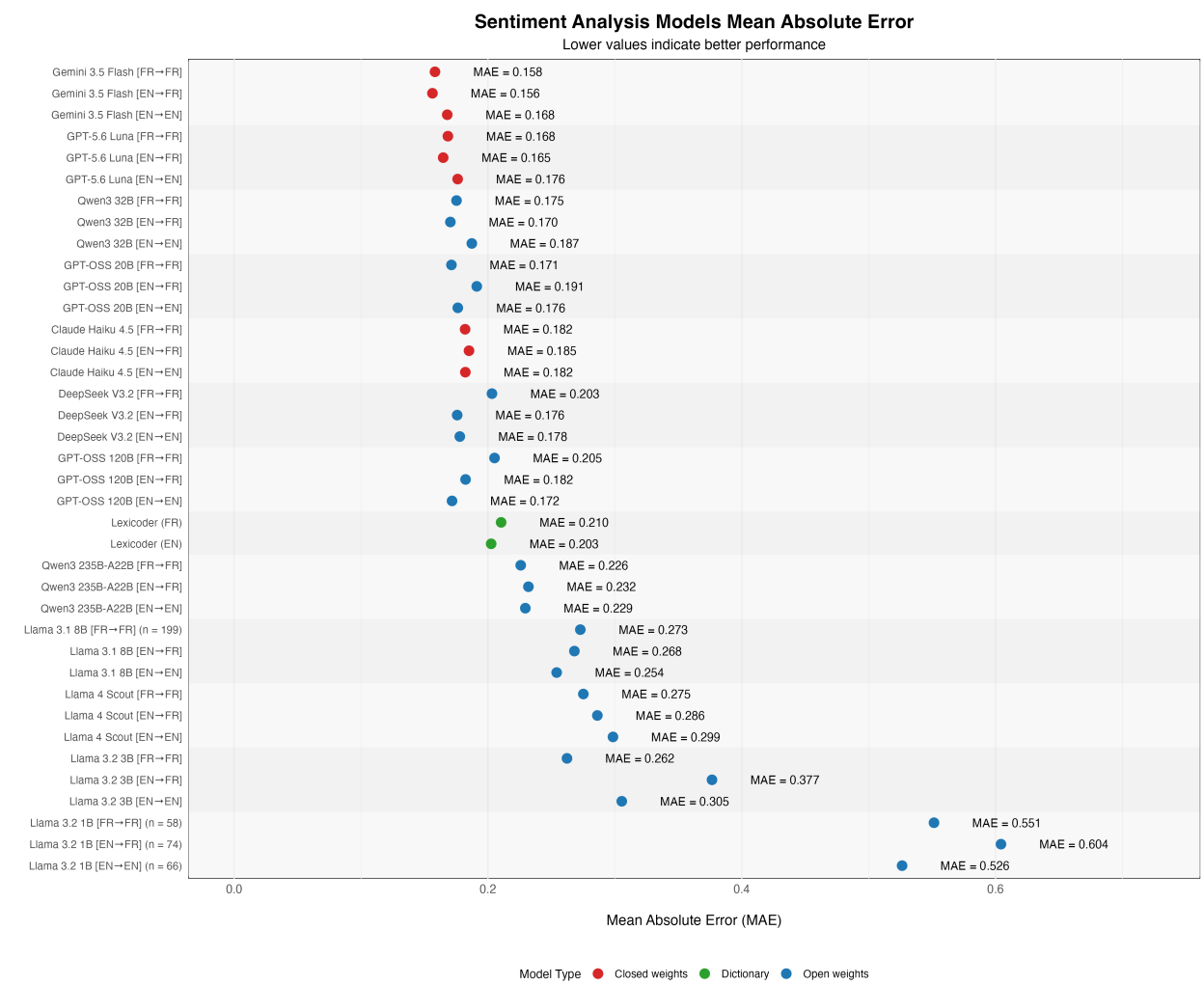


Figure 2. Cross-lingual sentiment analysis performance by Mean Absolute Error. Models grouped by type (color) and sorted by MAE (lower is better). Prompting mechanisms within clusters ordered as FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text).

Figure 4 shows the mean absolute error (MAE) across models and conditions.

Figure B2
Sentiment Analysis Comparison Over Time

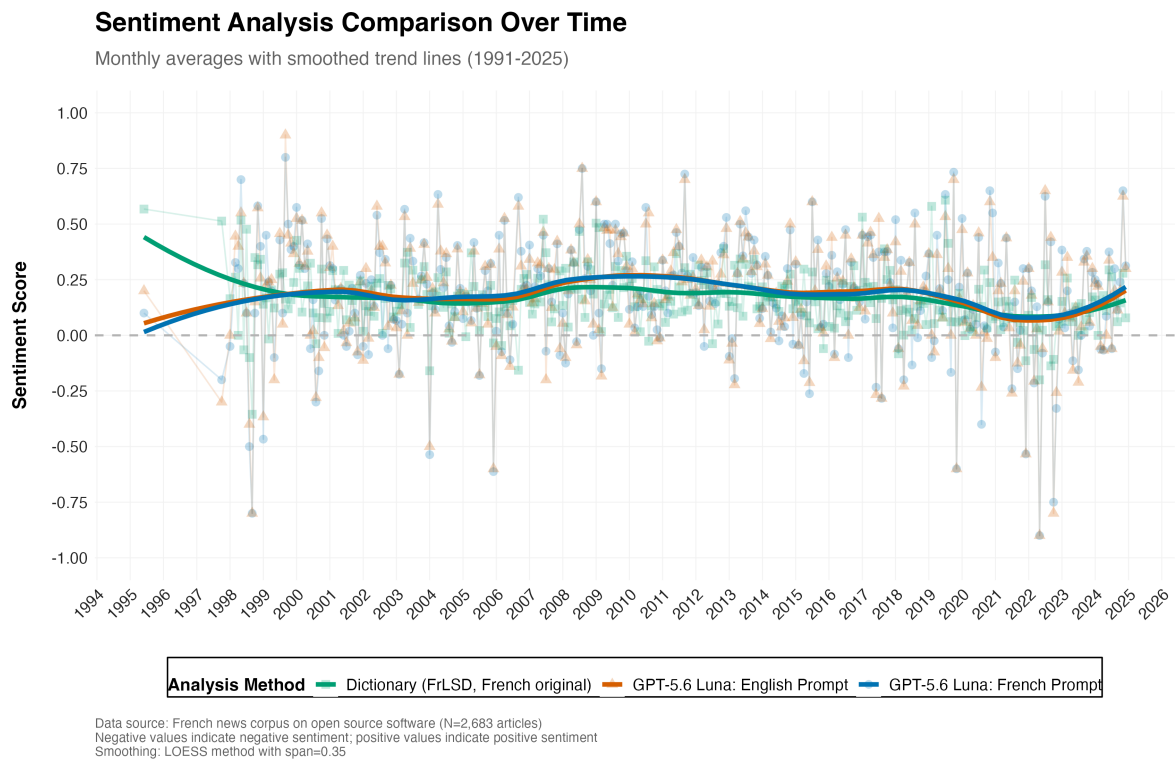


Figure 5 shows monthly mean sentiment for the 2,683 articles, scored by GPT-5.6 Luna with French prompts and with English prompts, and by the French Lexicoder dictionary on the original text. Points are monthly means and curves are LOESS smooths. The two prompt languages produce nearly identical series ($r = 0.97$ across 322 months). The dictionary agrees with the model only weakly ($r = 0.42$ across months, 0.36 across articles). The smoothed curves look alike because none of the three series shows a strong trend, not because the dictionary and the model agree on individual months.

Table B1

Cost of Scoring 1,000 Sentences at List Prices (Upper Bound; Cached Prompt Tokens Counted at the Full Input Rate; Input Tokens and Cost Not Reported for Llama 3.2 1B, Whose Endpoint Returned a Constant Token Count)

Model	Provider	Weights	API calls per usable score	Input tokens	Output tokens	USD per 1,000 sentences
GPT-4 OSS 20B	OpenRouter	Open	1.02	657,496	87,892	0.018
Llama 3.2 3B	OpenRouter	Open	1.00	595,142	7,320	0.018
GPT-4 OSS 120B	OpenRouter	Open	1.00	667,425	95,660	0.020
Qwen3 32B	OpenRouter	Open	1.00	592,086	5,696	0.027
Qwen3 235B-A22B	OpenRouter	Open	1.00	584,886	7,752	0.030
Llama 4 Scout	OpenRouter	Open	1.00	564,947	8,467	0.059
DeepSeek V3.2	OpenRouter	Open	1.00	559,594	9,340	0.067
Llama 3.1 8B	OpenRouter	Open	1.01	593,398	8,062	0.074
GPT-5.6 Luna	OpenAI	Closed	1.00	552,225	51,819	0.096
Claude Haiku 4.5	Anthropic	Closed	1.00	648,705	12,600	0.395
Gemini 3.5 Flash	Google	Closed	1.03	568,160	658,571	3.766
Llama 3.2 1B	OpenRouter	Open	22.68	—	168,789	—

Table B2

Mean Input Tokens per API Call by Language Condition, and Tokens per Word by Text Language (Valid Calls Only; Llama 3.2 1B Excluded Because Its Endpoint Reported a Constant Token Count on Every Call)

Model	EN→EN	EN→FR	FR→FR	French-prompt overhead	Tokens per French word	Tokens per English word
GPT- OSS 120B	352.0	357.7	402.7	+45 (+12.6%)	1.42	1.19
GPT- OSS 20B	339.0	344.7	389.7	+45 (+13.1%)	1.42	1.19
Llama 4	294.1	299.9	343.9	+44 (+14.7%)	1.41	1.19
Scout						
GPT- 5.6	288.0	293.7	338.7	+45 (+15.3%)	1.42	1.19
Luna						
Gemini 3.5	285.0	292.1	337.1	+45 (+15.4%)	1.46	1.17
Flash						
Qwen3 32B	296.5	307.7	382.7	+75 (+24.4%)	1.69	1.21
DeepSeek V3.2	280.1	290.3	361.3	+71 (+24.5%)	1.61	1.19
Qwen3 235B- A22B	292.5	303.7	378.7	+75 (+24.7%)	1.69	1.21
Llama 3.2 3B	295.2	306.8	388.8	+82 (+26.7%)	1.69	1.20
Llama 3.1 8B	295.2	306.8	389.0	+82 (+26.8%)	1.69	1.20
Claude Haiku 4.5	313.8	328.2	439.2	+111 (+33.8%)	1.87	1.25

Table B3*F1 Scores for 3-Category Classification*

Model	negative	neutral	positive	weighted_f1
Claude Haiku 4.5 (FR→FR)	0.762	0.375	0.832	0.778
GPT-5.6 Luna (FR→FR)	0.748	0.385	0.8	0.755
GPT-5.6 Luna (EN→FR)	0.73	0.364	0.776	0.733
Claude Haiku 4.5 (EN→FR)	0.69	0.273	0.778	0.715
DeepSeek V3.2 (FR→FR)	0.672	0.216	0.765	0.697
GPT-5.6 Luna (EN→EN)	0.688	0.367	0.736	0.695
Qwen3 235B-A22B (FR→FR)	0.752	0.243	0.716	0.695
DeepSeek V3.2 (EN→FR)	0.696	0.226	0.737	0.688
GPT-OSS 120B (FR→FR)	0.721	0.314	0.712	0.687
Llama 4 Scout (FR→FR)	0.643	0.291	0.747	0.682
Qwen3 235B-A22B (EN→FR)	0.719	0.254	0.706	0.679
Llama 4 Scout (EN→FR)	0.667	0.258	0.732	0.678
GPT-OSS 120B (EN→FR)	0.717	0.324	0.689	0.672
Claude Haiku 4.5 (EN→EN)	0.678	0.256	0.715	0.671
GPT-OSS 20B (EN→FR)	0.7	0.268	0.677	0.655
GPT-OSS 20B (FR→FR)	0.691	0.258	0.68	0.654
Llama 4 Scout (EN→EN)	0.647	0.246	0.686	0.643
Gemini 3.5 Flash (FR→FR)	0.655	0.264	0.673	0.639
DeepSeek V3.2 (EN→EN)	0.672	0.258	0.657	0.633
Llama 3.1 8B (EN→FR)	0.652	0.281	0.646	0.623
Llama 3.2 3B (FR→FR)	0.465	0.195	0.747	0.62
Llama 3.1 8B (EN→EN)	0.627	0.265	0.653	0.618
Gemini 3.5 Flash (EN→FR)	0.642	0.242	0.646	0.616
Llama 3.1 8B (FR→FR)	0.615	0.174	0.643	0.604
Gemini 3.5 Flash (EN→EN)	0.667	0.233	0.613	0.603
GPT-OSS 20B (EN→EN)	0.614	0.234	0.635	0.601
Qwen3 235B-A22B (EN→EN)	0.698	0.227	0.591	0.6
Lexicoder (EN)	0.574	0.231	0.636	0.588
GPT-OSS 120B (EN→EN)	0.678	0.218	0.552	0.569
Qwen3 32B (EN→FR)	0.554	0.165	0.621	0.568
Lexicoder (FR)	0.474	0.289	0.631	0.557
Qwen3 32B (FR→FR)	0.515	0.177	0.611	0.55
Llama 3.2 3B (EN→EN)	0.167	0.261	0.73	0.52
Qwen3 32B (EN→EN)	0.559	0.19	0.52	0.509
Llama 3.2 3B (EN→FR)	0	0.222	0.757	0.481
Llama 3.2 1B (FR→FR)	0	0.138	0.648	0.443
Llama 3.2 1B (EN→FR)	0	0	0.568	0.338
Llama 3.2 1B (EN→EN)	0	0.15	0.457	0.286

Table B4*F1 Scores for 7-Category Classification*

Model	very_neg	negative	somewhat_neg	neutral	somewhat_pos	positive	very_pos	weighted
Claude Haiku 4.5 (FR→FR)	0.4	0.68	0.455	0.375	0.54	0.612	0	0.537
GPT-5.6 Luna (FR→FR)	0.286	0.5	0.393	0.385	0.538	0.462	0.133	0.479
GPT-5.6 Luna (EN→EN)	0.364	0.5	0.338	0.367	0.51	0.542	0.25	0.47
GPT-5.6 Luna (EN→FR)	0.333	0.5	0.258	0.364	0.564	0.407	0.125	0.455
DeepSeek V3.2 (EN→EN)	0.333	0.533	0.419	0.258	0.415	0.588	0.667	0.448
Claude Haiku 4.5 (EN→EN)	0.286	0.642	0.4	0.256	0.409	0.568	0	0.448
GPT-OSS 120B (EN→FR)	0.333	0.566	0.4	0.324	0.416	0.485	0.133	0.434
DeepSeek V3.2 (EN→FR)	0.444	0.667	0.429	0.226	0.349	0.493	0	0.416
DeepSeek V3.2 (FR→FR)	0.333	0.368	0.469	0.216	0.4	0.527	0.167	0.415
Claude Haiku 4.5 (EN→FR)	0.5	0.68	0.364	0.273	0.347	0.511	0.2	0.413
GPT-OSS 20B (EN→FR)	0.235	0.431	0.269	0.268	0.37	0.535	0	0.376
GPT-OSS 120B (EN→EN)	0.444	0.692	0.281	0.218	0.288	0.536	0.286	0.373
Gemini 3.5 Flash (FR→FR)	0.222	0.583	0.25	0.264	0.31	0.571	0.2	0.37
GPT-OSS 120B (FR→FR)	0.308	0.545	0.333	0.314	0.319	0.447	0.154	0.369
GPT-OSS 20B (FR→FR)	0.364	0.6	0.286	0.258	0.325	0.448	0.286	0.367
GPT-OSS 20B (EN→EN)	0.4	0.528	0.235	0.234	0.338	0.508	0	0.362
Gemini 3.5 Flash (EN→FR)	0.4	0.694	0.255	0.242	0.263	0.485	0.167	0.349
Qwen3 32B (EN→FR)	0.222	0.512	0.163	0.165	0.286	0.615	0	0.334
Qwen3 235B-A22B (FR→FR)	0.19	0.407	0.32	0.243	0.319	0.369	0.087	0.33
Gemini 3.5 Flash (EN→EN)	0.364	0.652	0.222	0.233	0.23	0.5	0	0.324
Qwen3 32B (FR→FR)	0.333	0.558	0.125	0.177	0.236	0.632	0	0.315
Qwen3 235B-A22B (EN→FR)	0.182	0.321	0.24	0.254	0.35	0.339	0.091	0.315
Qwen3 235B-A22B (EN→EN)	0.182	0.275	0.302	0.227	0.279	0.408	0.133	0.299
Llama 3.1 8B (EN→FR)	0.111	0.12	0.192	0.281	0.319	0.507	0.2	0.297
Lexicoder (FR)	0	0	0.253	0.289	0.479	0.057	0	0.29
Llama 4 Scout (FR→FR)	0.125	0.167	0.182	0.291	0.341	0.356	0.1	0.286
Lexicoder (EN)	0	0	0.267	0.231	0.472	0	0	0.276
Llama 3.1 8B (FR→FR)	0.143	0.197	0.179	0.174	0.239	0.543	0	0.267
Llama 3.2 3B (FR→FR)	0	0.364	0.2	0.195	0.24	0.395	0	0.266
Qwen3 32B (EN→EN)	0.222	0.364	0.172	0.19	0.168	0.567	0.333	0.261
Llama 3.1 8B (EN→EN)	0.138	0.222	0.136	0.265	0.246	0.479	0.143	0.261
Llama 4 Scout (EN→FR)	0.154	0.261	0.143	0.258	0.281	0.329	0.08	0.257
Llama 4 Scout (EN→EN)	0.105	0.28	0.125	0.246	0.234	0.434	0.154	0.252